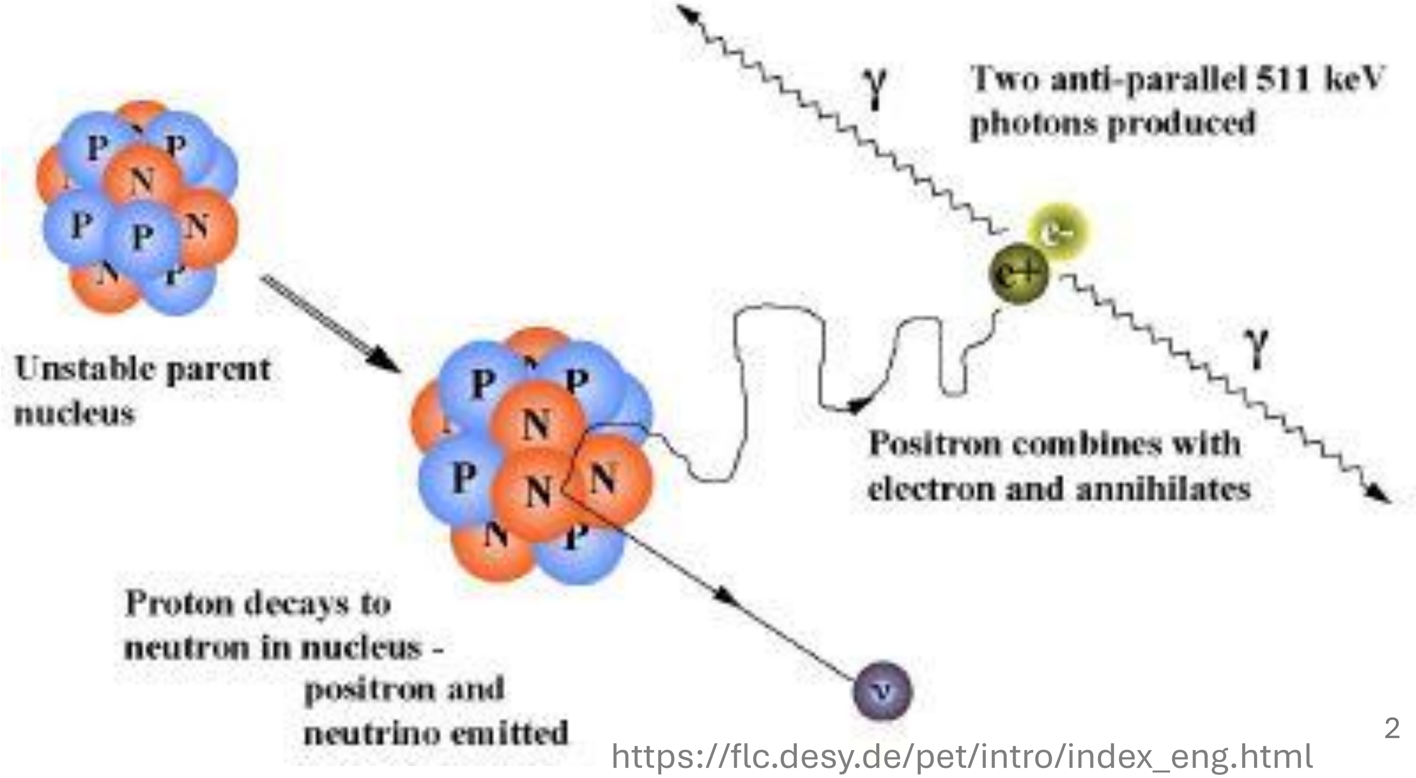
Impact of positron range on image reconstruction in ^{18}F and ^{44}Sc PET imaging

Anna Giebułtowska

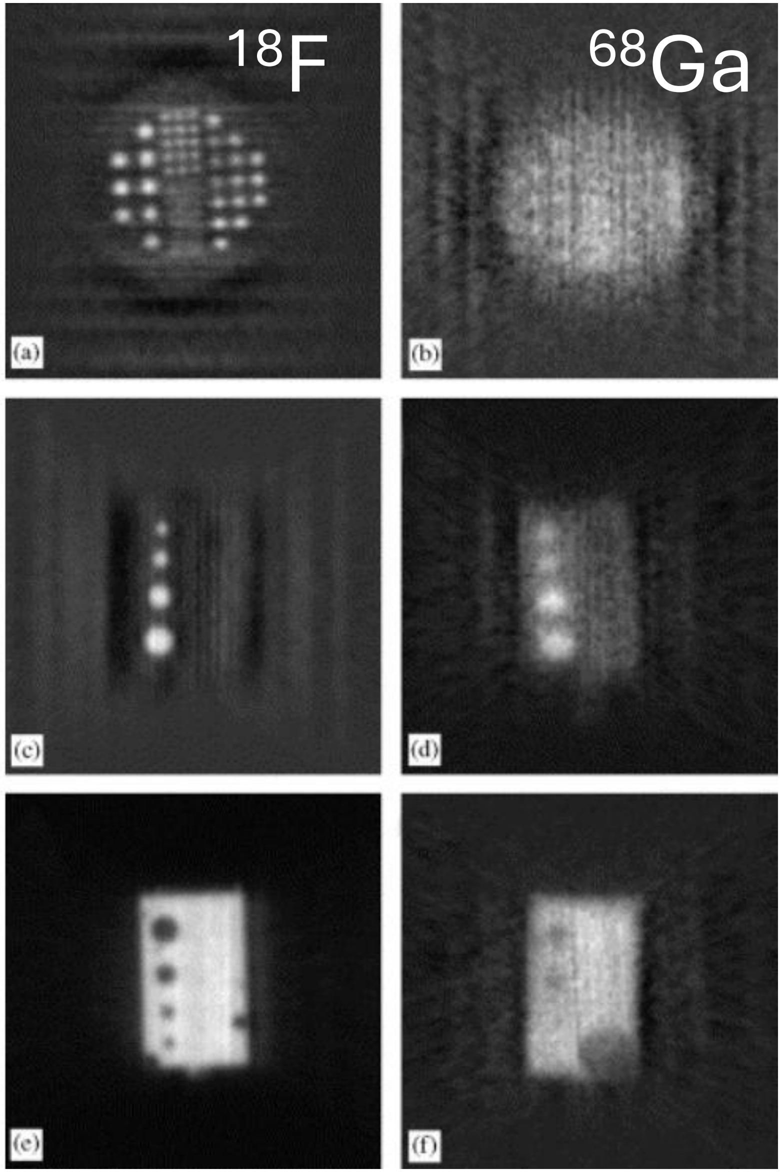
Agenda

- Motivation
- Current stage of work
- Summary
- Future research



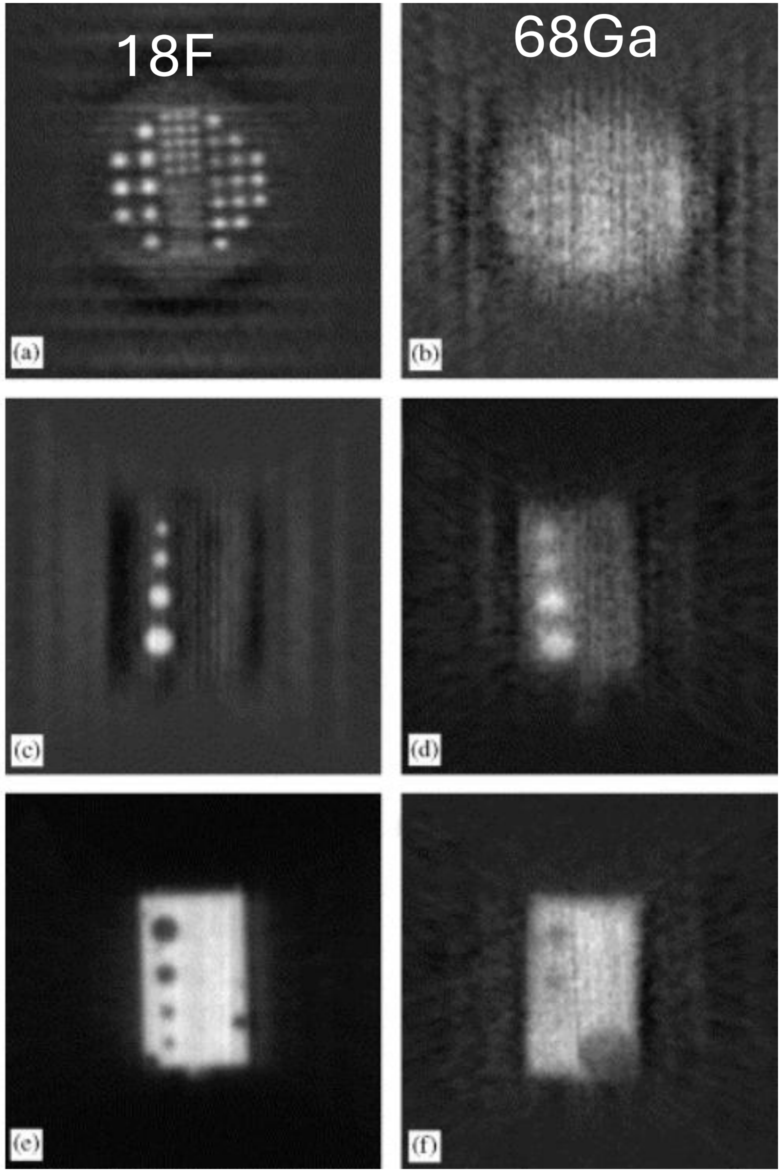
https://flc.desy.de/pet/intro/index_eng.html

Motivation



Partige:
https://www.sciencedirect.com/science/article/pii/S0168900206016640

Motivation



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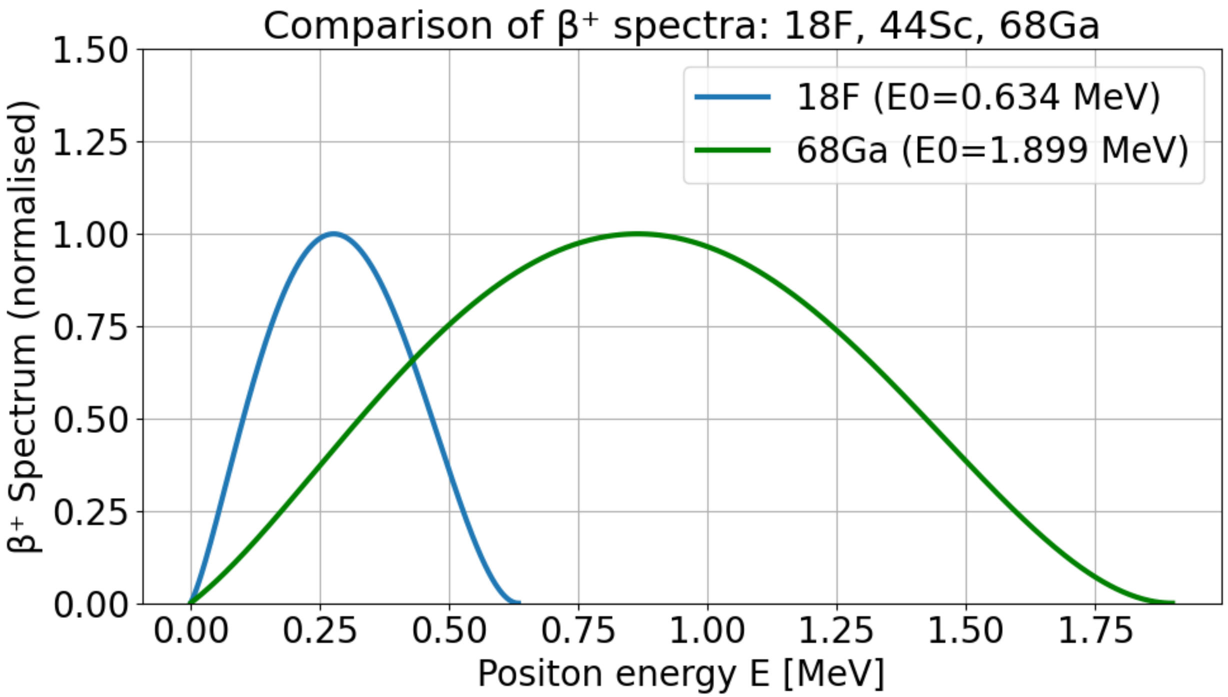
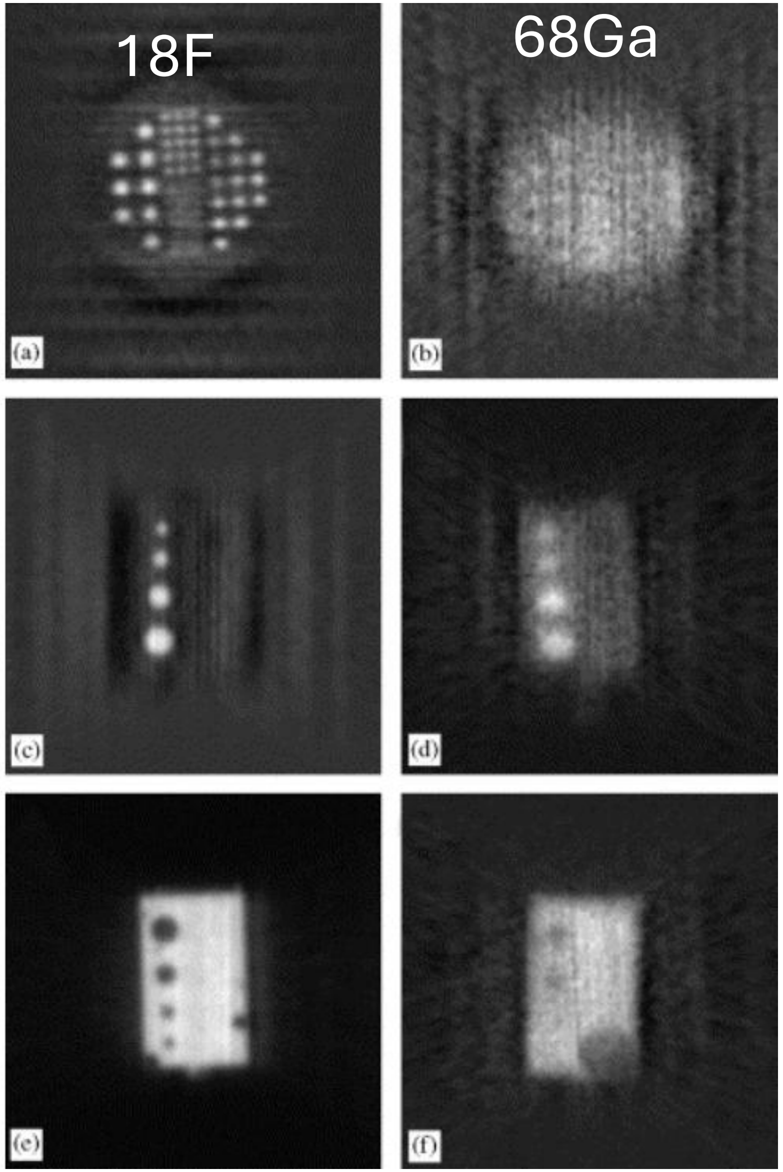
https://www.sciencedirect.com/science/article/pii/S0168900206016640



Motivation



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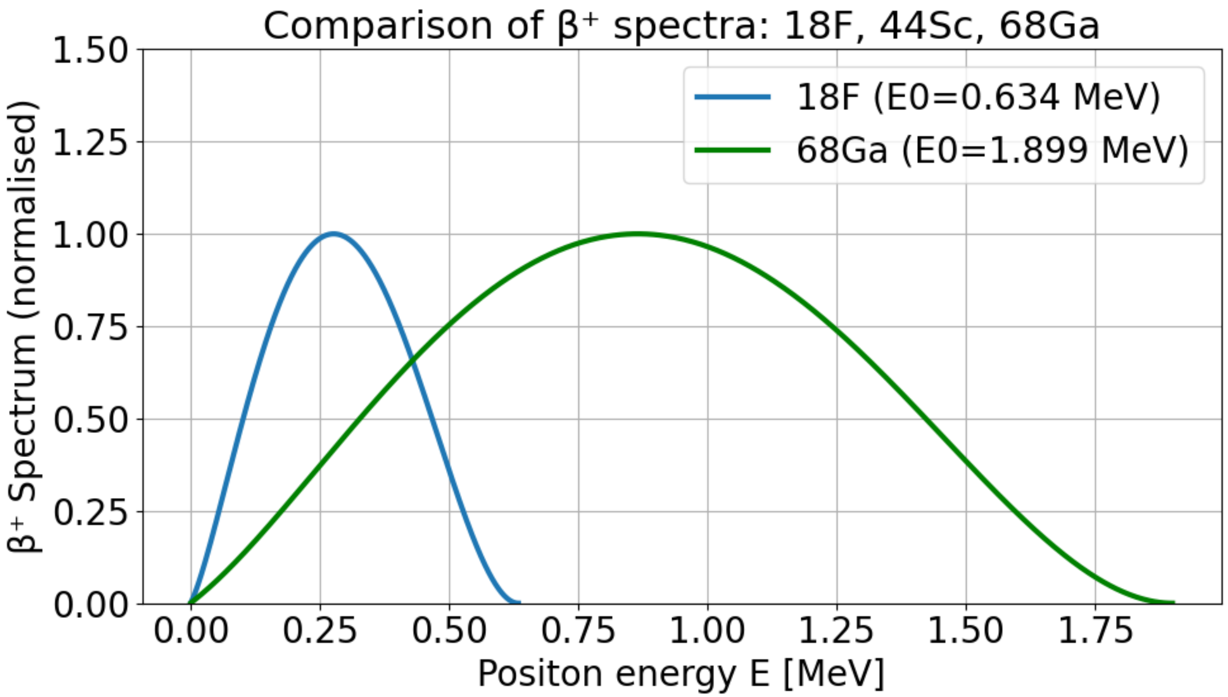
Levin CS, Hoffman EJ. Calculation of positron range and its effect on PET resolution. Phys Med Biol.



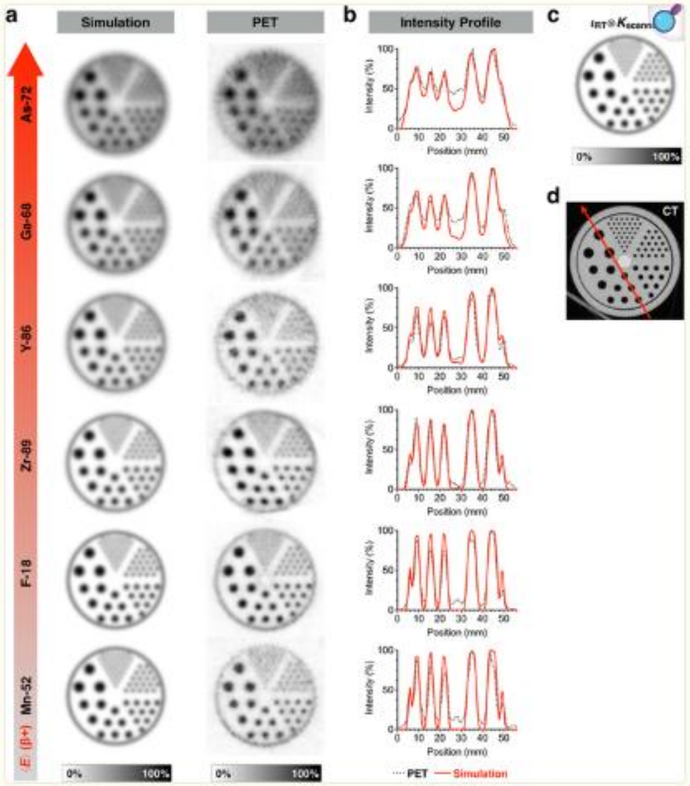
$$N(E)\,dE = g\,F(Z,E)\,pE(E_{\max} - E)^2\,dE$$

$$F_{\text{allowed}}(Z,E) = \frac{2\pi\eta}{1 - e^{-2\pi\eta}}$$

$$\eta = -Z\alpha E/p$$

Motivation

Fig. 5.



Open in a new tab

a Comparison of PET images and simulations of the Data Spectrum® Micro Jaszczak phantom with different positron emitters. Data are normalized to the maximum pixel intensity in each respective image. **b** Intensity profiles taken over line indicated in **d**. **c** Simulated distribution of radiotracer (*i.e.*, zero positron range blur) convolved with $PSF_{scanner}$ represents upper limit on achievable resolution. **d** Computed tomography (CT) scan of the unfilled Jaszczak phantom.

Carter: https://pmc.ncbi.nlm.nih.gov/articles/PMC6805144/#TFN1

MC underestimates!

Motivation

Fig. 5.

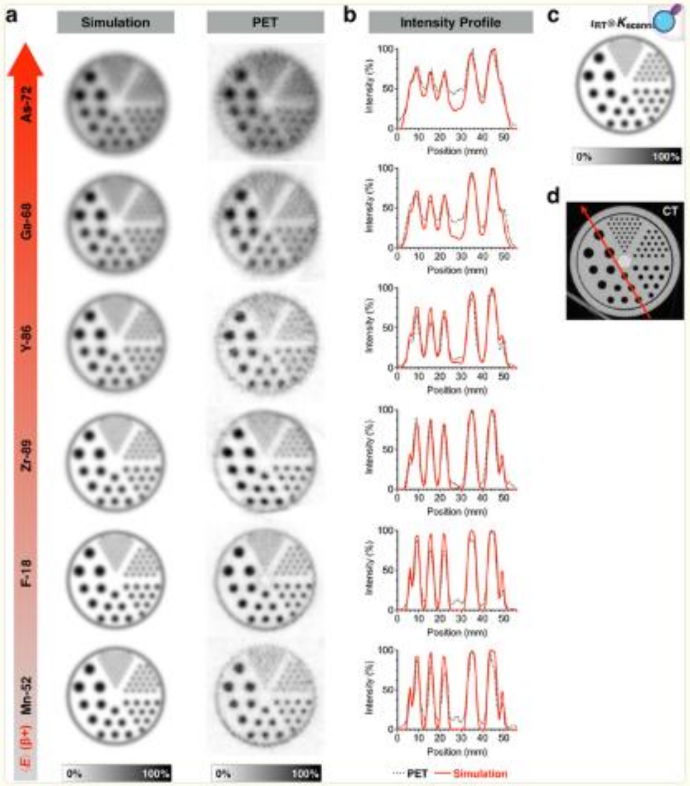


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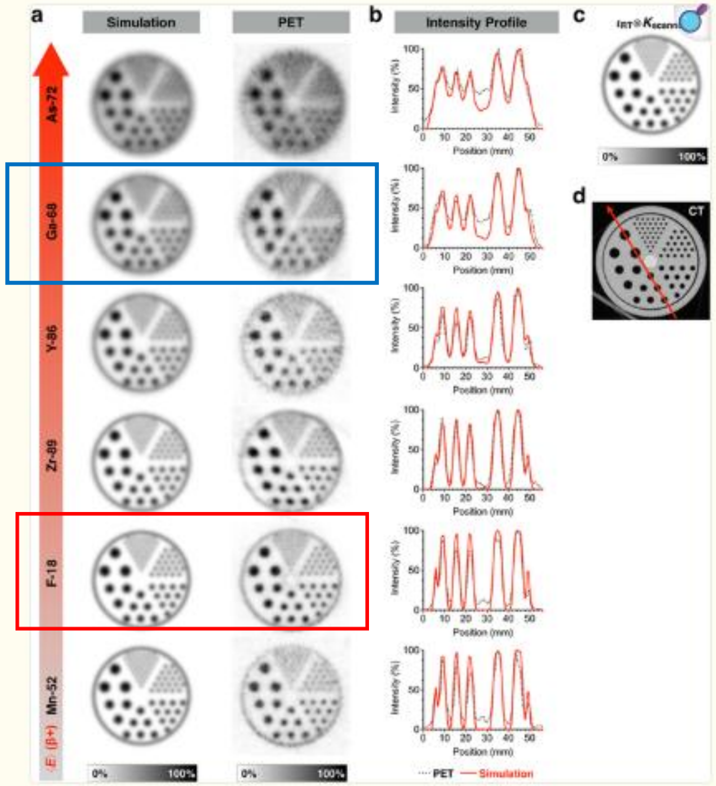
Table 4. Positron ranges in water for each radioisotope studied. Comparison between our Monte Carlo results (in bold) and existing values taken from different sources: for R_{mean}: Partridge *et al* (2006) (reference (1)), Bailey *et al* (2003) (reference (2)), Comtat (private communication) (reference (3)) and Defrise and Trebossen (private communication) (reference (4)); for R_{max}: Evans (1972) (reference (5)). Experimental measurements are taken from Derenzo (1993) and Derenzo *et al* (1930) (reference (6)) and from Cho *et al* (1975) (reference (7)). For all these values, relative disagreement with our result is reported into brackets.

Isotopes	R_{mean} (mm) Theory	R_{mean} (mm) Experimental	R_{max} (mm) Theory
^{18}F	**0.66** 0.6[1] (9.1%) 0.6[2] (9.1%) 0.5437[3] (17.6%) 0.6[4] (9.1%)	0.54[6] (18.2%) 0.9[7] (36.1%)	**2.633** 2.3[5] (12.6%)
^{11}C	**1.266** 1.1[1] (13.1%) 1.1[2] (13.1%) 1.12[4] (11.5%)	0.92[6] (27.3%) 1.095[7] (13.5%)	**4.456** 3.9[5] (12.5%)
^{13}N	**1.730** 1.5[1] (13.3%) 1.5[2] (13.3%) 1.44[4] (16.8%)	1.39[7] (19.6%)	**5.752** 5.1[5] (11.3%)
^{15}O	**2.965** 2.5[1] (15.7%) 2.5[2] (15.7%) 2.261[3] (23.8%) 2.22[4] (25.1%)	1.785[7] (39.8%)	**9.132** 8[5] (12.4%)
^{68}Ga	**3.559** 2.9[1] (18.5%) 2.4[4] 32.6%)	2.8[6] (21.3%) 1.975[7] (44.5%)	**10.273** 9[6] (12.4%)
^{82}Rb	**7.491** 5.9[1] (21.2%) 4.7[4] (37.2%)	6.1[6] (18.6%) 2.9[7] (61.2%)	**18.603** 18[5] (3.2%)

Champion: https://iopscience.iop.org/article/10.1088/0031-9155/52/22/004/pdf

Motivation

Fig. 5.



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a Comparison of PET images and simulations of the Data Spectrum® Micro Jaszczak phantom with different positron emitters. Data are normalized to the maximum pixel intensity in each respective image. b Intensity profiles taken over line indicated in d. c Simulated distribution of radiotracer (*i.e.*, zero positron range blur) convolved with $PSF_{scanner}$ represents upper limit on achievable resolution. d Computed tomography (CT) scan of the unfilled Jaszczak phantom.

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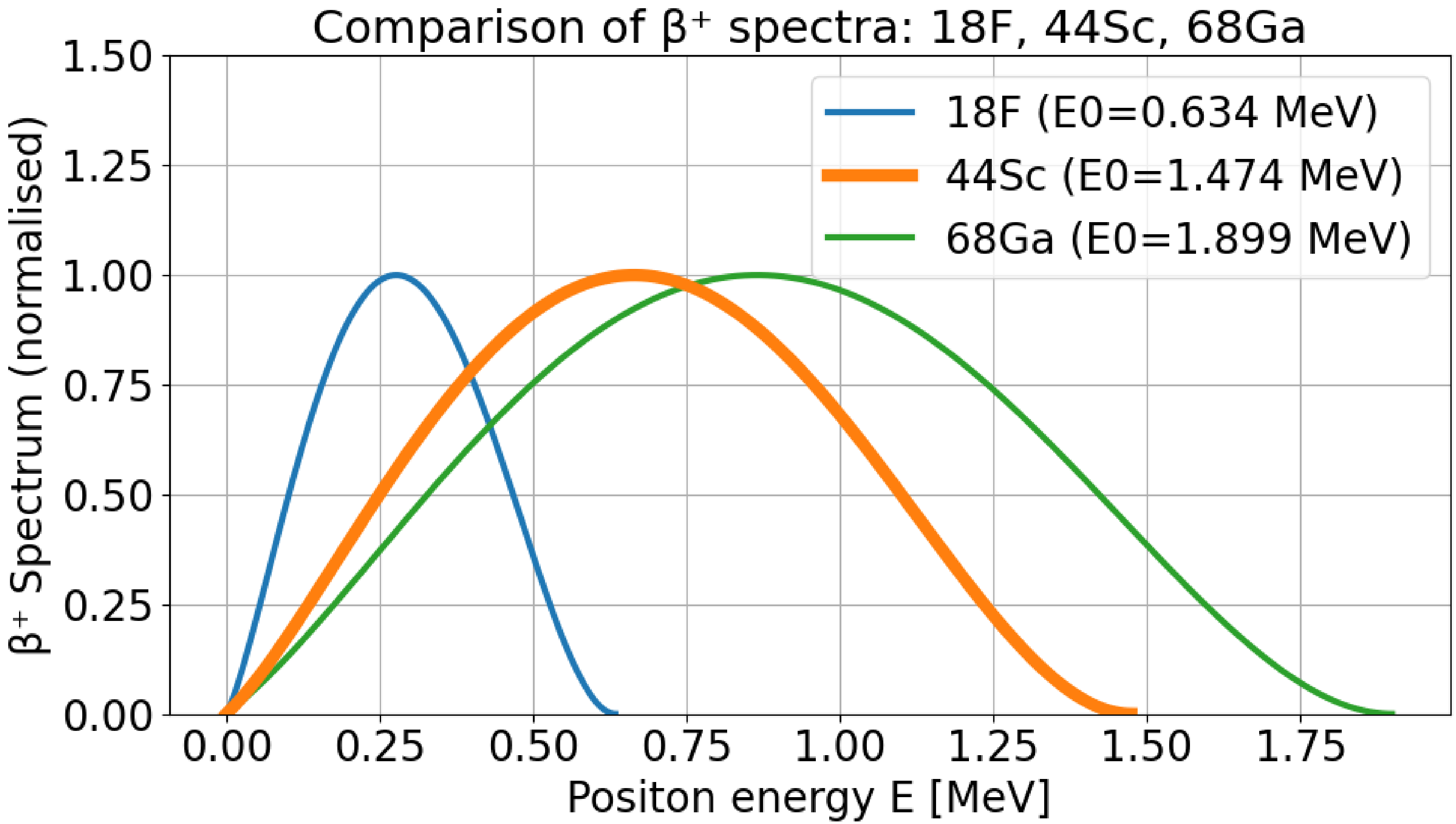
	R_{mean} (mm)		R_{max} (mm)
Isotopes	Theory	Experimental	Theory
^{18}F	**0.66** $0.6^{(1)}$ (9.1%) $0.6^{(2)}$ (9.1%) $0.5437^{(3)}$ (17.6%) $0.6^{(4)}$ (9.1%)	$0.54^{(6)}$ (18.2%) $0.9^{(7)}$ (36.1%)	**2.633** $2.3^{(5)}$ (12.6%)
^{11}C	**1.266** $1.1^{(1)}$ (13.1%) $1.1^{(2)}$ (13.1%) $1.12^{(4)}$ (11.5%)	$0.92^{(6)}$ (27.3%) $1.095^{(7)}$ (13.5%)	**4.456** $3.9^{(5)}$ (12.5%)
^{13}N	**1.730** $1.5^{(1)}$ (13.3%) $1.5^{(2)}$ (13.3%) $1.44^{(4)}$ (16.8%)	$1.39^{(7)}$ (19.6%)	**5.752** $5.1^{(5)}$ (11.3%)
^{15}O	**2.965** $2.5^{(1)}$ (15.7%) $2.5^{(2)}$ (15.7%) $2.261^{(3)}$ (23.8%) $2.22^{(4)}$ (25.1%)	$1.785^{(7)}$ (39.8%)	**9.132** $8^{(5)}$ (12.4%)
^{68}Ga	**3.559** $2.9^{(1)}$ (18.5%) $2.4^{(4)}$ 32.6%)	$2.8^{(6)}$ (21.3%) $1.975^{(7)}$ (44.5%)	**10.273** $9^{(6)}$ (12.4%)
^{82}Rb	**7.491** $5.9^{(1)}$ (21.2%) $4.7^{(4)}$ (37.2%)	$6.1^{(6)}$ (18.6%) $2.9^{(7)}$ (61.2%)	**18.603** $18^{(5)}$ (3.2%)

Champion: https://iopscience.iop.org/article/10.1088/0031-9155/52/22/004/pdf

Energy spectrum



$$N(E)\,dE = g\,F(Z,E)\,pE(E_{\max}-E)^2\,dE$$

$$F_{\text{allowed}}(Z,E) = \frac{2\pi\eta}{1-e^{-2\pi\eta}}$$

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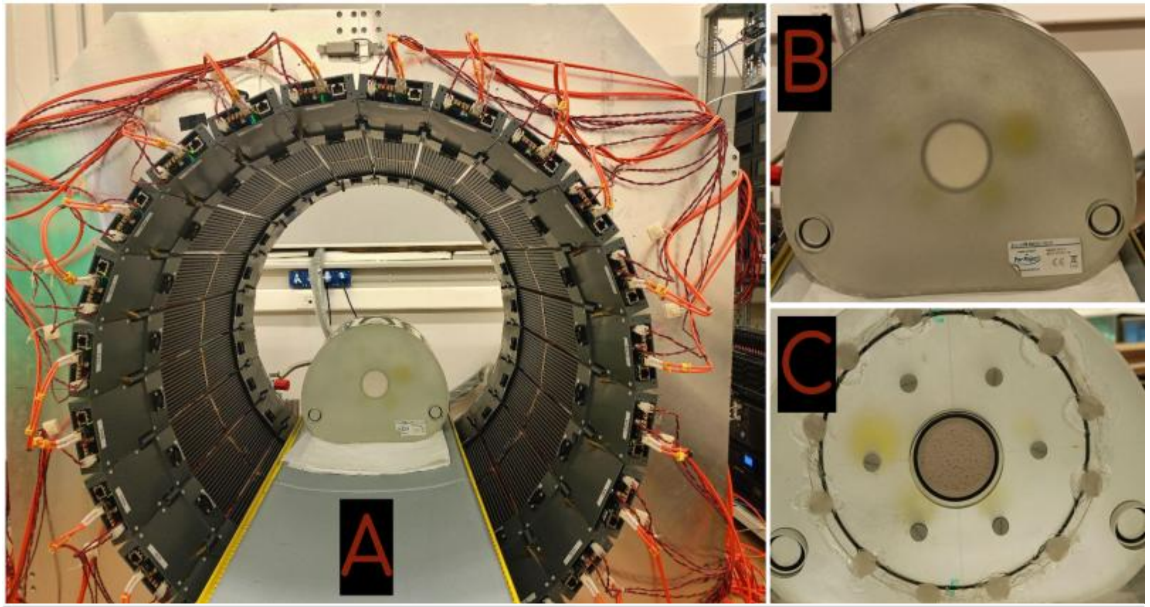
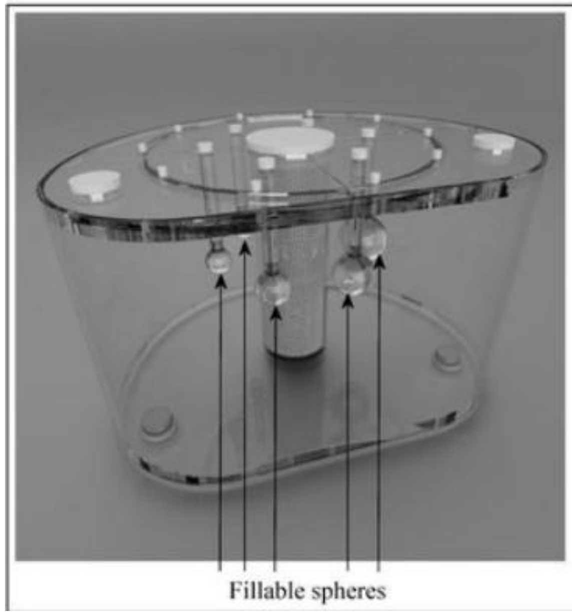
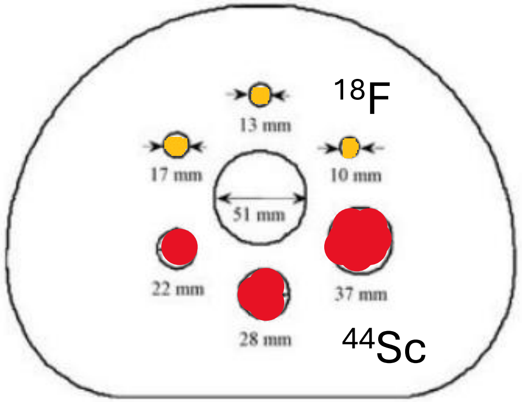
Experiment



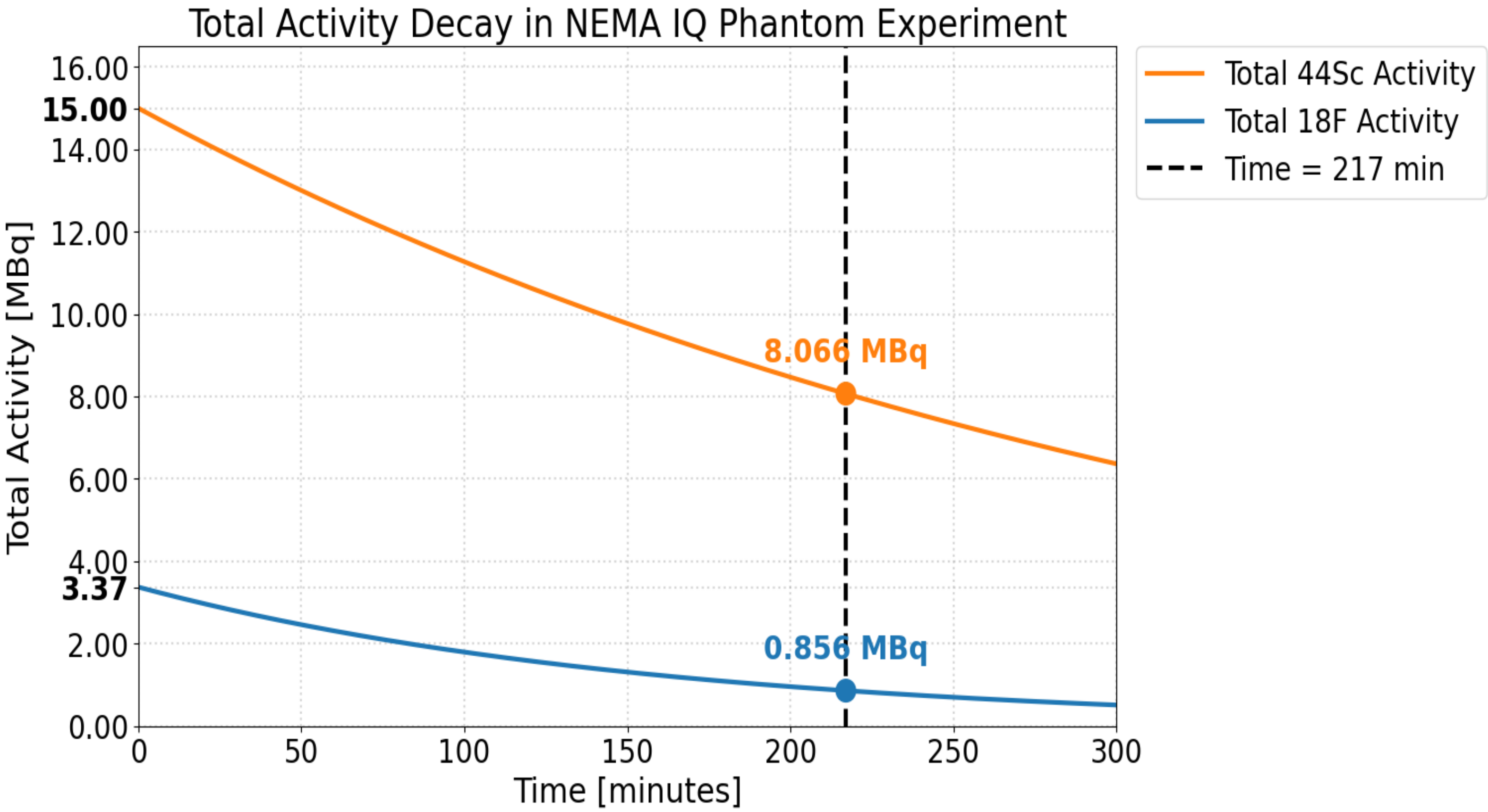




Raport_Scandium_exp2025
section 3.4.4

Activity decay

$^{44}Sc/^{18}F =$ 4,45104



$^{44}Sc/^{18}F =$ 9,42290

$$A = A_0 e^{-\lambda t} \quad \lambda = \frac{\ln(2)}{T_{1/2}}$$

$$T_{1/2}^{Sc} = 242.52\, min$$

$$T_{1/2}^{F} = 109.73\, min$$

doi.org/10.1016/j.apradiso.2022.110507

Table with 44Sc

Radionuclide	Energy_mean [MeV]
Mn-52	0,242
F-18	**0,25**
Cu-64	0,28
C-11	0,39
Zr-89	0,4
N-13	0,49
Sc-44	**0,63**
Y-86	0,66
O-15	0,74
I-129	0,82

https://www.nupecc.org/pub/npmed2014.pdf

Table with 44Sc

Radionuclide	Energy_mean [MeV]	Range_Lit [mm]
Mn-52	0,242	0,59
F-18	**0,25**	**0,6**
Cu-64	0,28	0,7
C-11	0,39	1,1
Zr-89	0,4	1,1
N-13	0,49	1,5
Sc-44	**0,63**	**2**
Y-86	0,66	2,1
O-15	0,74	2,4
I-129	0,82	2,9

https://www.nupecc.org/pub/npmed2014.pdf

Table with 44Sc

Radionuclide	Energy_mean [MeV]	Range_Lit [mm]	Range_calulated [mm]
Mn-52	0,242	0,59	0,561
F-18	**0,25**	**0,6**	**0,590**
Cu-64	0,28	0,7	0,701
C-11	0,39	1,1	1,145
Zr-89	0,4	1,1	1,188
N-13	0,49	1,5	1,586
Sc-44	**0,63**	**2**	**2,245**
Y-86	0,66	2,1	2,391
O-15	0,74	2,4	2,787
I-129	0,82	2,9	3,190

$$r = 0.412 \cdot E^{(1.27-0.0954\cdot\ln E)}$$

$$X = \frac{r}{\rho}$$

https://koza.if.uj.edu.pl/files/9f8966edebfcd92b0d62916d3801a6f4/praca_doktorska_pawlik1.pdf#page=4.48&gsr=0

https://www.nupecc.org/pub/npmed2014.pdf

Table with 44Sc

Radionuclide	Energy_mean [MeV]	Range_Lit [mm]	Range_calulated [mm]	Difference in Range
Mn-52	0,242	0,59	0,561	-4,92%
F-18	**0,25**	**0,6**	**0,590**	**-1,71%**
Cu-64	0,28	0,7	0,701	0,13%
C-11	0,39	1,1	1,145	4,09%
Zr-89	0,4	1,1	1,188	7,98%
N-13	0,49	1,5	1,586	5,75%
Sc-44	**0,63**	**2**	**2,245**	**12,25%**
Y-86	0,66	2,1	2,391	13,85%
O-15	0,74	2,4	2,787	16,11%
I-129	0,82	2,9	3,190	10,00%

$$r = 0.412 \cdot E^{(1.27 - 0.0954 \cdot \ln E)}$$

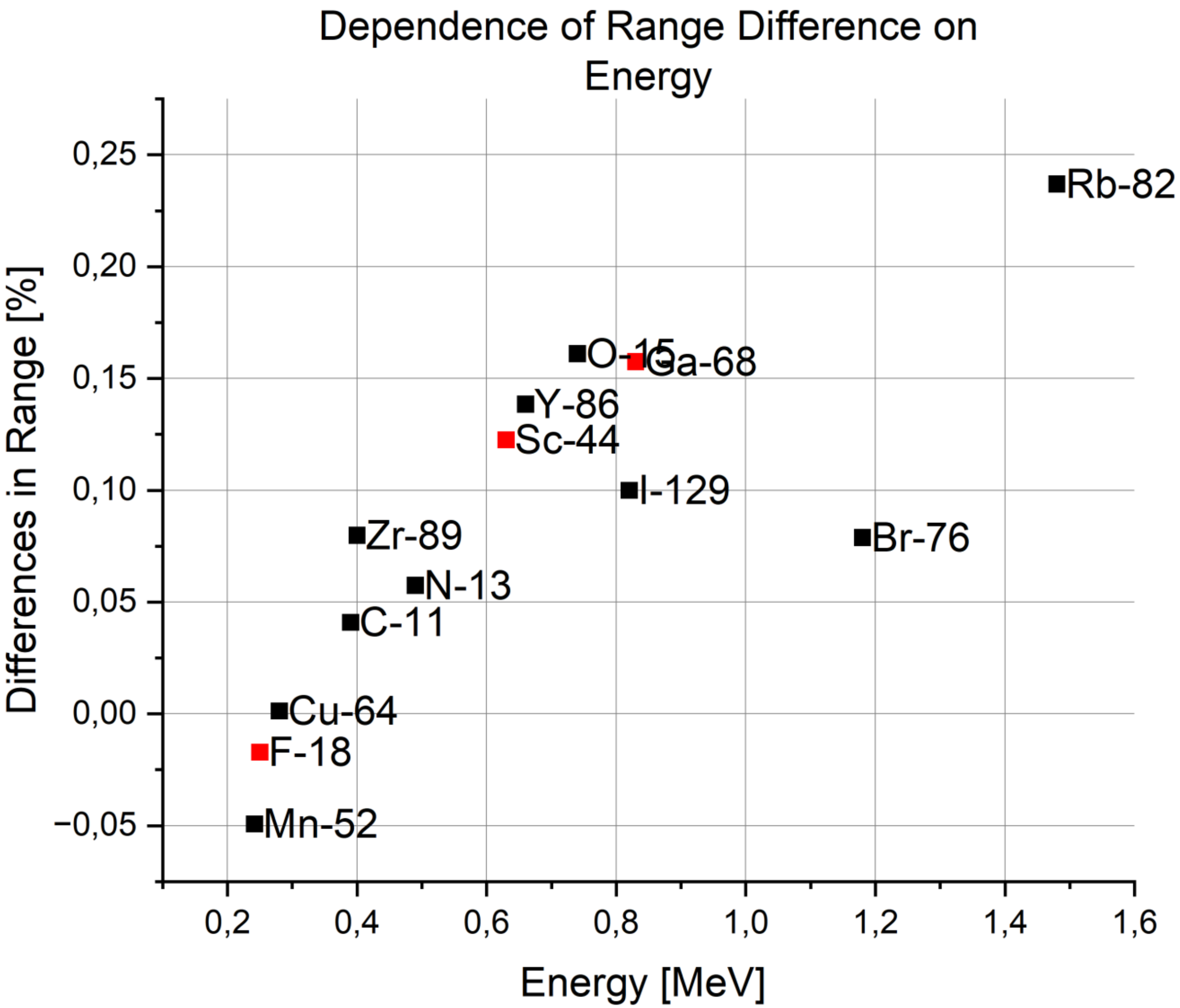
$$X = \frac{r}{\rho}$$

$$\text{Difference in range} = \frac{\text{Range\_calculated} - \text{Range\_Lit}}{\text{Range\_Lit}}$$

https://koza.if.uj.edu.pl/files/9f8966edebfcd92b0d62916d3801a6f4/praca_doktorska_pawlik1.pdf#page=4.48&gsr=0

https://www.nupecc.org/pub/npmed2014.pdf

Range Difference vs. Energy



Radionuclide	Energy_mean [MeV]	Range_Lit [mm]	Range_calulated [mm]	Difference in Range
F-18	**0,25**	**0,6**	**0,590**	**-1,71%**
Sc-44	**0,63**	**2**	**2,245**	**12,25%**
Ga-68	**0,83**	**2,8**	**3,241**	**15,75%**

Summary

- We observe that a positron's energy affects image quality in PET imaging
- The current research is inconclusive when it comes to ranges of positrons with higher energies – the ones that affect blurring of a PET image the most
- ^{44}Sc as an emerging PET radioisotope is lacking research on PET image quality dependant on positron range

Future research

Literature

vs

Analytical

vs

Experiment

Thank You for Your attention